



Flaw in Solar Panels

After 40 years of searching for efficient methods, scientists have identified a flaw in solar panel efficiency. <https://digit.in/july19-45>



Mole Rat immune to Wasabi

A type of African mole rat is immune to the pain caused by the allyl isothiocyanate (AITC) in wasabi. <https://digit.in/july19-44>

3D printing human tissue without scaffolding

Scaffolds used in 3D printing human tissues can be problematic. A group of scientists find another way.

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D printing of human tissue and organs seemed to be a farfetched idea a few years ago, even when

3D printing was being used more and more frequently. However, now we have established methods of actually printing human tissue involving scaffolds. However, this method doesn't seem to be the most reliable one. Scientists have actually found a way to 3D print human tissue and, subsequently, organs without using this process. This newly-developed technique overcomes some of the shortcomings of the scaffold-technique by using hydrogel instead.

TRADITIONAL METHOD AND ITS PROBLEMS

3D printing of biological tissue (including organs), so far, was usually accomplished using microscale scaffolds. Human cells are seeded into a material with a scaffolding kind of microstructure. This structure basically becomes a three-dimensional home to the cells within which they nest, grow and



Scaffolds are traditionally used to 3D print tissues and organs

then subsequently reproduce. At some point, the cells take over completely as the host materials, which the scaffold is made out of, biodegrades. In this way, nothing remains apart from the biotissue which has been 3D printed in the desired form. However, according to a fair number of scientists, especially the ones researching this process at the University of Illinois at Chicago, this process, of using scaffolding to obtain 3D printed human tissue and organs does have its fair share of drawbacks.

Firstly, it is very tricky to get the timing of the biodegradation of the scaffolds just right since this decomposition process must coincide exactly with the maturation of the tissue or the organ. Secondly, the biodegradation of the material used in the scaffolds for this process can actually produce by-products of a toxic nature. Lastly, the scaffolds can also interfere with the development of cell-to-cell communication which is extremely essential to ensure the proper formation of the 3D printed tissue. These set of problems, along with the desire to do something radical in this field, caused a group of scientists to experiment and actually